

YONATAN BASON

MECHANICAL ENGINEER

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SUMMARY

Mechanical engineer with strong foundations in structural mechanics, thermal systems, and mechanical design. Experienced in component and assembly-level development, hardware integration, and performance validation. Proven ability to translate specifications into manufacturable designs, perform structural and thermal evaluations, and support testing through iterative design refinement.

HIGHLIGHTS OF EXPERTISE

- Solid Mechanics & Structural Analysis
- Stress & Deformation Evaluation
- High-Pressure Systems (Up to 40 bar)
- Mechanical Assembly & Validation
- Failure Analysis
- CAD Modeling (SolidWorks, CREO)
- DFM & GD&T
- Finite Element Fundamentals
- Experimental Design & Testing
- Cross-Functional Collaboration

PROFESSIONAL EXPERIENCE

Arlo Industries

Engineer

March 2026 –

San Francisco, California

- Lead R&D on complex hardware nodes, overseeing the end-to-end design, simulation, and integration of sensor-heavy platforms.
- Utilize CAD software for parametric modeling of high-stress mechanical assemblies and custom enclosures.
- Apply material science and 3D printing expertise to engineer functional, field-tested components using industrial-grade materials.
- Execute full-stack development, coordinating mechanical build requirements with software implementation to meet mission-critical tracking benchmarks.

Technion Rocketry Club

Propulsion Engineer

Jun 2022 – Jan 2025

Haifa, Israel

- Designed and modeled hybrid rocket propulsion assemblies in SolidWorks, developing components, subassemblies, and structural support hardware operating at chamber pressures up to 40 bar.
- Performed analytical structural and thermal evaluations (hoop stress, axial stress, thermal expansion) to verify design margins and requirements adherence.
- Generated detailed part and assembly drawings to support fabrication and integration.
- Participated in hardware builds, integration, and static fire testing, evaluating system performance and recommending iterative design modifications.
- Improved stiffness-to-weight ratio and structural robustness through design refinement based on test feedback.

Faculty of Mechanical Engineering, Technion

Undergraduate Researcher

Dec 2022 – Jan 2025

Haifa, Israel

- Performed full-field strain and deformation analysis using Digital Image Correlation (DIC) under dynamic loading conditions.
- Correlated experimental strain data with solid mechanics models to validate structural behavior and quantify load redistribution.
- Identified a 14% increase in horizontal force redirection through vector-based deformation analysis, improving understanding of load path behavior.
- Designed and documented custom mechanical fixtures to improve boundary condition repeatability and structural test consistency.
- Documented experimental setup, analytical methodology, and performance results to support reproducibility and future testing phases.

Alexion Pharmaceuticals (Rare Disease Unit of AstraZeneca; acquired LogicBio in 2022)

Engineering Intern (3 Terms)

2021 – 2023

Lexington, MA

- Designed and prototyped mechanical hardware for a cost-efficient purification system, reducing process costs by 3× through structured design iteration and validation testing.
- Supported engineering scale-up of bioprocess systems (2L–50L), evaluating structural integrity, thermal performance, and fluid handling behavior under increased operational demands.
- Participated in equipment installation, performance verification, and troubleshooting, recommending mechanical design modifications to improve reliability and repeatability.
- Collaborated with cross-functional teams to generate documentation including procedures, test records, and updated design specifications.
- Improved assay throughput by 20% through thermal workflow optimization and systematic bottleneck analysis.

EDUCATION

Technion Israel Institute of Technology
B.Sc. Mechanical Engineering (Mechatronics, Robotics)
– **Relevant Coursework:** Machine Design, Solid Mechanics, Dynamics, Heat Transfer, Thermodynamics, Fluid Mechanics, CAD

Haifa, Israel
2020 – 2025

TECHNICAL SKILLS

- **Software:** MATLAB, Python, SolidWorks, CREO
- **Manufacturing:** CNC Machining, 3D Printing, Injection Molding
- **Languages:** English, Hebrew